

Hon'ble Chief Minister of Karnataka Shri Siddaramaiah unveiling the Karnataka Quantum Roadmap at the inaugural session of the Quantum India Bengaluru Summit 2025.

KARNATAKA'S QUANTUM LEAP: FROM INNOVATION HUB TO GLOBAL QUANTUM POWERHOUSE

Bengaluru, long celebrated as India's Silicon Valley, is now scripting a new chapter in the country's technological journey. The successful hosting of the Quantum India Bengaluru Summit 2025 has catapulted Karnataka to the forefront of the global quantum technology landscape. The two-day event, held on July 31 and August 1, was not just a conference—it was a declaration of intent: Karnataka is ready to lead the world into the quantum era.

By ARUN KUMAR HV

A Historic Beginning

The summit, inaugurated by **Chief Minister Siddaramaiah** and **Deputy Chief Minister D.K. Shivakumar,** brought together an unprecedented gathering of **1,951 delegates** from **19 countries.** It featured **Nobel Laureates Prof. Duncan Haldane and Prof. David Gross,** global experts, policymakers, and industry leaders. Over 24 sessions, 75 speakers engaged in in-depth deliberations on quantum computing, cryptography, sensing, AI, and their applications in sectors ranging from healthcare to defense.

"Karnataka has long been India's innovation capital. Now, we are taking a quantum leap towards becoming a global leader," said **Minister for Science & Technology N.S. Boseraju,** the driving force behind this initiative.

The Rise of Q-City

One of the most ambitious projects announced is the establishment of **Q-City** near Bengaluru—a futuristic hub integrating academic institutions, R&D centers, manufacturing clusters, and data centers. "Q-City will do for quantum what Bengaluru once did for IT," said Minister Boseraju. The city will also host a **Quantum Hardware Park** and the country's first to manufacture quantum processors and components.

KARNATAKA QUANTUM ROAD MAP:

4-Phase Quantum Roadmap!

During the inaugural ceremony of Quantum India Bengaluru Summit, Hon'ble Chief Minister Siddaramaiah unveiled the **Karnataka Quantum Roadmap,** a strategic blueprint aimed at positioning the state as a global quantum leader by 2035.

The roadmap focuses on:

- Phase 1: Foundation & Capacity Building
- Phase 2: R&D & Pilot Deployment
- Phase 3: Manufacturing & Commercialization
- Phase 4: Global Quantum Powerhouse

The **Government of Karnataka** will soon establish a **Quantum Task Force** to implement the Karnataka Quantum Roadmap and provide guidance on policy interventions and strategic collaborations.

Major Announcements from Quantum India Bengaluru 2025

- **₹1,000 crore** Karnataka Quantum Mission Fund for infrastructure, R&D, startups, and skilling.
- Quantum Innovation Fund to nurture **100+ startups** in quantum technology.
- **Q-City** (Quantum City) to be established near Bengaluru with a Hardware Park, FabLine, and Innovation Hub.
- Quantum Skilling Programs in **20+ Colleges,** Support for **150 Phd** Fellowships.
- Indigenous development of **1,000-qubit** processing capacity.
- Target: **$20 billion** quantum economy and **20%** share of global market by **2035.**

Global Partnerships, Local Strength

The summit also served as a platform for forging international collaborations. Countries including **Germany, Japan, Israel, Singapore, and the USA** expressed keen interest in partnering with Karnataka on quantum ventures. A notable highlight was **the MoU between IIIT Dharwad, Raichur institutions, and QpiAI** to establish the **Q-Vidya 8-qubit quantum computer**—a critical step towards indigenous capacity building.

Building a Quantum Workforce

Recognizing that talent is the backbone of innovation, Karnataka announced **quantum skilling programs** in over **20 colleges,** with plans to scale further. Through the **STREAM Labs initiative**, quantum